

A Self-Supervised Framework for Automated Mouse Brain Extraction in Data-Constrained MRI Studies

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Abstract—The objective of this study is to develop an efficient framework for 2D mouse brain extraction in MRI using self-supervised learning (SSL) on unlabeled datasets. We hypothesize that SSL enables the model to learn meaningful anatomical representations that significantly improve brain extraction performance when limited labeled data are available. We used a in-house mouse brain dataset consisting of 21 subjects with varying numbers of 2D slices per subject. We compare a standard UNet architecture trained from scratch with a two stage UNet-style SSL framework wherein in the first stage, the encoder was pretrained using masked image modeling (MIM) on the unlabeled training images divided into patches with randomly masking and then the pretrained encoder was fine-tuned for supervised segmentation using the same training settings as the baseline UNet. Performance was evaluated using L1 loss for reconstruction during the first stage and the Dice similarity coefficient (DSC) for the precision of brain extraction. We evaluated multiple combinations of patch sizes (8, 16, 32) and masking ratios (0.4 to 0.9) during the SSL pretraining stage. A patch size of 32×32 with a masking ratio of 0.5 achieved the lowest reconstruction error, which is the most effective learning of image representations. Compared with a baseline UNet trained from scratch under the same labeling budget, the model trained after SSL-pre-training improves the DSC from 0.80 to 0.88 and yields qualitatively sharper boundary delineation and improved robustness. The experimental results suggest that SSL pretraining can improve 2D mouse brain segmentation in data-constrained environments.

Index Terms—Self-supervised learning, masked image modeling, medical image segmentation, mouse brain imaging, deep learning, UNet, neuroimaging

I. INTRODUCTION

Accurate brain segmentation in small animal imaging is a fundamental step in many preclinical neuroscience workflows [1], including morphometric analysis, atlas registration, volumetric quantification, and disease model comparisons. In neuroimaging in mouse, the quality of the segmentation directly affects the reliability of downstream measurements such as regional brain volume, structural deformation, and lesion burden. Despite steady progress in deep learning-based segmentation [2], a persistent limitation in preclinical settings is the scarcity of high-quality pixel wise annotations. Manual labeling of mouse brain slices is time-consuming, requires

specialized expertise, and becomes increasingly impractical as datasets expand across subjects, protocols, and acquisition sites.

Convolutional neural networks (CNNs), particularly encoder and decoder architectures such as UNet [3], have become a standard approach for biomedical segmentation because they preserve fine spatial details using skip connections and multiscale feature extraction. Large-scale benchmarking studies have further demonstrated the effectiveness of UNet-based frameworks in medical imaging tasks [4]–[7]. However, UNet and similar supervised models typically require substantial labeled datasets to generalize reliably. When trained under limited labels, segmentation models tend to overfit, show unstable boundaries, and exhibit reduced robustness to variation in contrast, noise, and anatomical differences across subjects. This motivates the adoption of learning strategies that can leverage unlabeled data more effectively.

Self-supervised learning (SSL) has emerged as an effective paradigm to reduce dependence on annotations by learning transferable feature representations from unlabeled data [8], [9]. Recent studies have also demonstrated the effectiveness of SSL in brain MRI analysis tasks [10]. In SSL, a model is trained using a pretext task that does not require manual labels, and the learned representation is then fine-tuned for the downstream supervised objective. Masked image modeling (MIM) is one of the most successful SSL strategies where a portion of image content is intentionally removed (masked), and the model learns to reconstruct the missing regions [11]. This forces the network to encode contextual and structural information, which can be especially beneficial in medical images where anatomical continuity and shape priors matter.

In this study, we develop and evaluate a UNet-based SSL segmentation framework for brain extraction (i.e., whole-brain segmentation) for mouse 2D slice in MRI. We compare a baseline UNet trained from scratch to a two-stage SSL approach where the encoder is first pretrained using a masked reconstruction task on unlabeled training images and then fine-tuned for supervised segmentation using labeled masks. We systematically evaluate masking strategies using different patch sizes and masking ratios and analyze their effects on

Stage one: Self-supervised learning

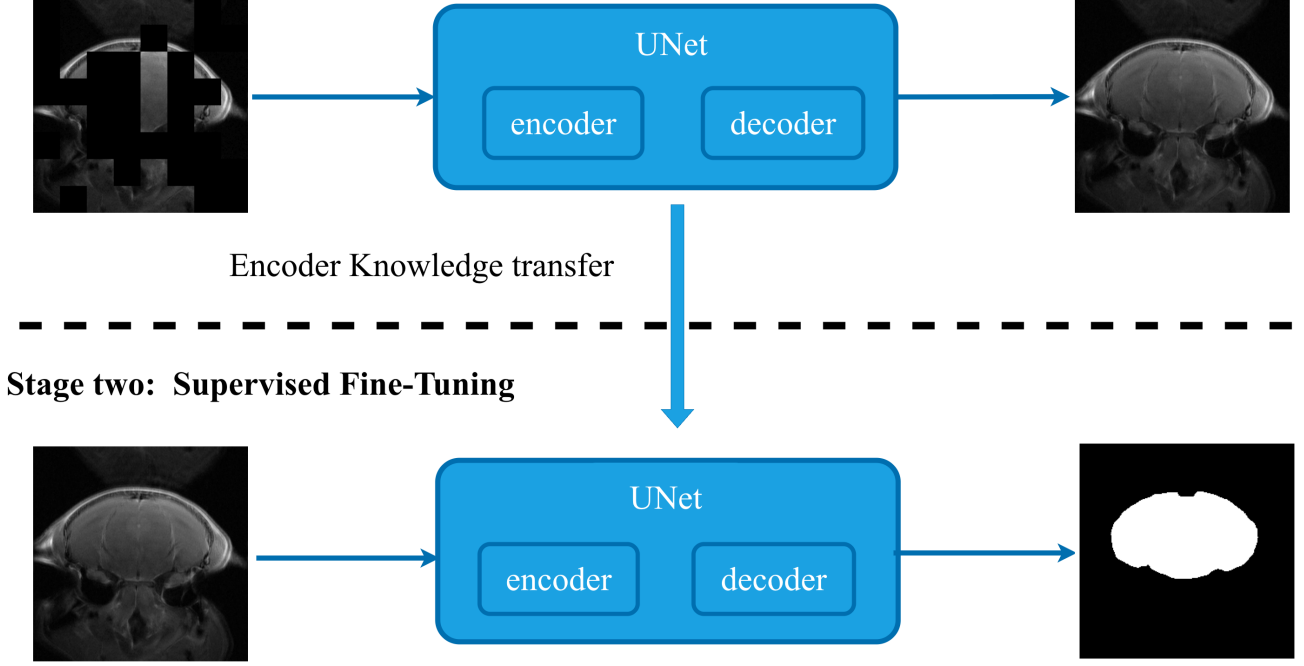


Fig. 1. Two-stage training strategy for mouse brain segmentation.

reconstruction loss during pretraining and Dice Similarity Coefficient (DSC) during segmentation. Our hypothesis is that SSL pretraining improves anatomical representation learning and produces more accurate and robust segmentation results under the same labeled data budget.

II. METHODS

A. Data Preprocessing

We used an in-house mouse brain dataset consisting of 20 subjects, where each subject contains a variable number of 2D slices with expert manual annotations. MR images were acquired using a Bruker 9.4T small animal MRI scanner. T1-weighted and T2-weighted images were obtained in the axial plane with an anteroposterior readout orientation, using a slice thickness of 1 mm, a field of view (FoV) of 14×14 mm, and a 90° flip angle. For T1-weighted imaging, the echo time (TE) and repetition time (TR) were 6.7 ms and 350 ms, respectively, whereas T2-weighted images were acquired with a TE of 10 ms and a TR of 2,500 ms.

To ensure a fair evaluation and avoid data leakage between anatomically similar slices from the same subject, we performed subject-level splitting and used a fixed split across all experiments. There were 14 subjects (152 slices) for training, 2 subjects (22 slices) for validation, and 4 subjects (44 slices) for testing. All images and masks were resized to 256×256 and converted to single-channel grayscale inputs. Data loading was implemented using a separate paired masked original dataset

class for the SSL pretraining stage and a paired image mask dataset class for supervised training. A fixed random seed was used for reproducibility.

We compare two approaches, a standard UNet architecture trained from scratch using labeled data [3] and a UNet-style self-supervised learning framework with a two-stage training strategy [12].

B. Standard UNet

For the first approach, input images were resized to 256×256 and trained with a batch size of 4. The network was trained upto 100 epochs by using the Adam optimizer [13] with an initial learning rate of $1e-3$ and a weight decay of $1e-4$. The learning rate was adaptively reduced by a factor of 0.5 after 2 epochs without improvement in validation loss by using the scheduler with a minimum learning rate of $1e-7$, and the model was optimized using binary cross-entropy with logits loss which is implemented in the PyTorch framework [14]. The validation loss was monitored at each epoch, the best-performing checkpoint was saved, and early stopping [15] was applied with patience of 15 epochs based on the validation loss.

C. UNet with Self-Supervised Learning

For the second approach, Fig. 1 illustrates the self-supervised learning (SSL) proposed two-stage training framework for mouse brain segmentation

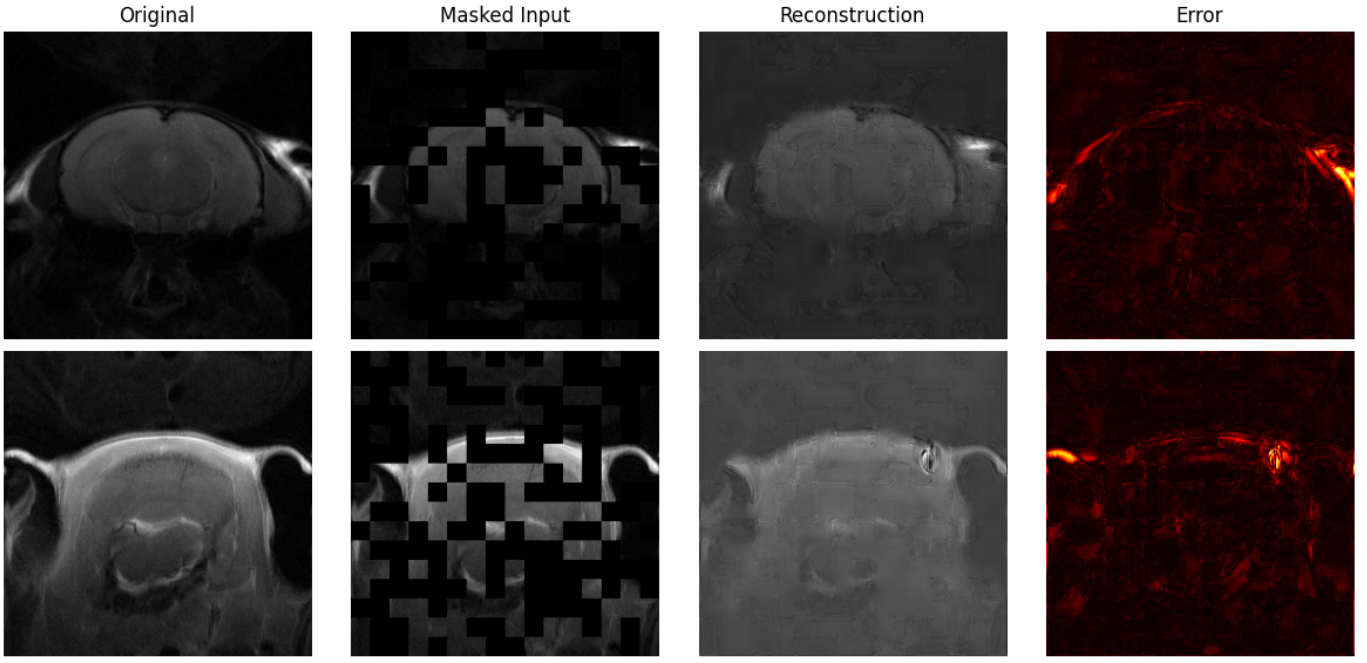


Fig. 2. Reconstruction performance during SSL pretraining. The encoder learns to recover masked regions using L_1 loss, demonstrating the model’s ability to recover masked regions and learn meaningful anatomical representations.

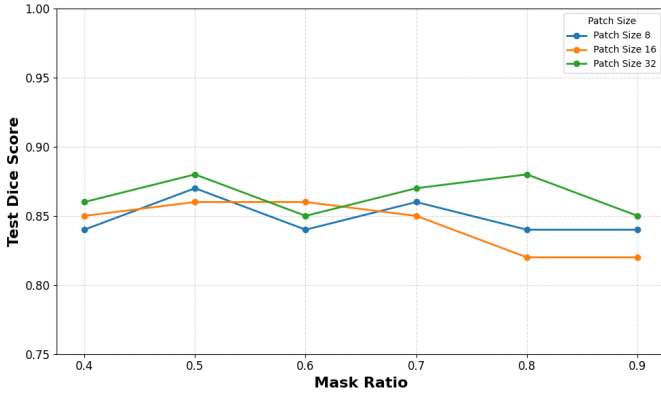


Fig. 3. Test Dice Similarity Coefficient score versus masking ratio for different patch sizes.

- In the first stage, the encoder was pretrained using Masked Image Modeling (MIM) on unlabeled training images. Each 256×256 image was divided into non-overlapping patches. We evaluated multiple combinations of patch sizes ($P \in \{8, 16, 32\}$) and masking ratios (ranging from 0.4 to 0.9, as a ratio of 1.0 would remove all information) to optimize pretraining performance. The model was tasked with reconstructing the original pixel values from the visible patches, thereby encouraging the encoder to learn robust anatomical and contextual representations [11]. The framework was optimized using an L_1 loss between the reconstructed output and the original image. Fig. 2 illustrates the reconstruction re-

sults, demonstrating the model’s ability to recover masked regions and capture meaningful structural features. The final configuration was selected based on the minimum reconstruction error across the hyperparameter grid. Pre-training was performed for up to 50 epochs by using the Adam optimizer [13] with an initial learning rate of $1e-4$ and a weight decay of $1e-5$. The learning rate was adaptively reduced using the scheduler [14] with a reduction factor of 0.5 and a minimum learning rate of $1e-7$.

- In the second stage, the pretrained encoder weights from the optimal SSL checkpoint were integrated into the segmentation architecture and fine-tuned for the supervised task. We maintained the same training hyperparameters as the baseline UNet to ensure a fair comparison. While pretraining performance was monitored via L_1 reconstruction loss, the final segmentation efficacy of both the baseline and fine-tuned models was evaluated using the Dice Similarity Coefficient (DSC) on the test set.

III. RESULTS

We evaluated the impact of varying patch sizes ($P \in \{8, 16, 32\}$) and masking ratios ($\rho \in [0.4, 0.9]$) during the SSL pretraining phase. As illustrated in Fig. 3, the optimal reconstruction performance was achieved with a patch size of 32×32 and a masking ratio of 0.5, yielding the minimum reconstruction error (L_1 loss = 0.0054). This configuration facilitated the most effective extraction of latent anatomical features for downstream tasks. For the supervised segmentation task, the baseline UNet trained from scratch achieved

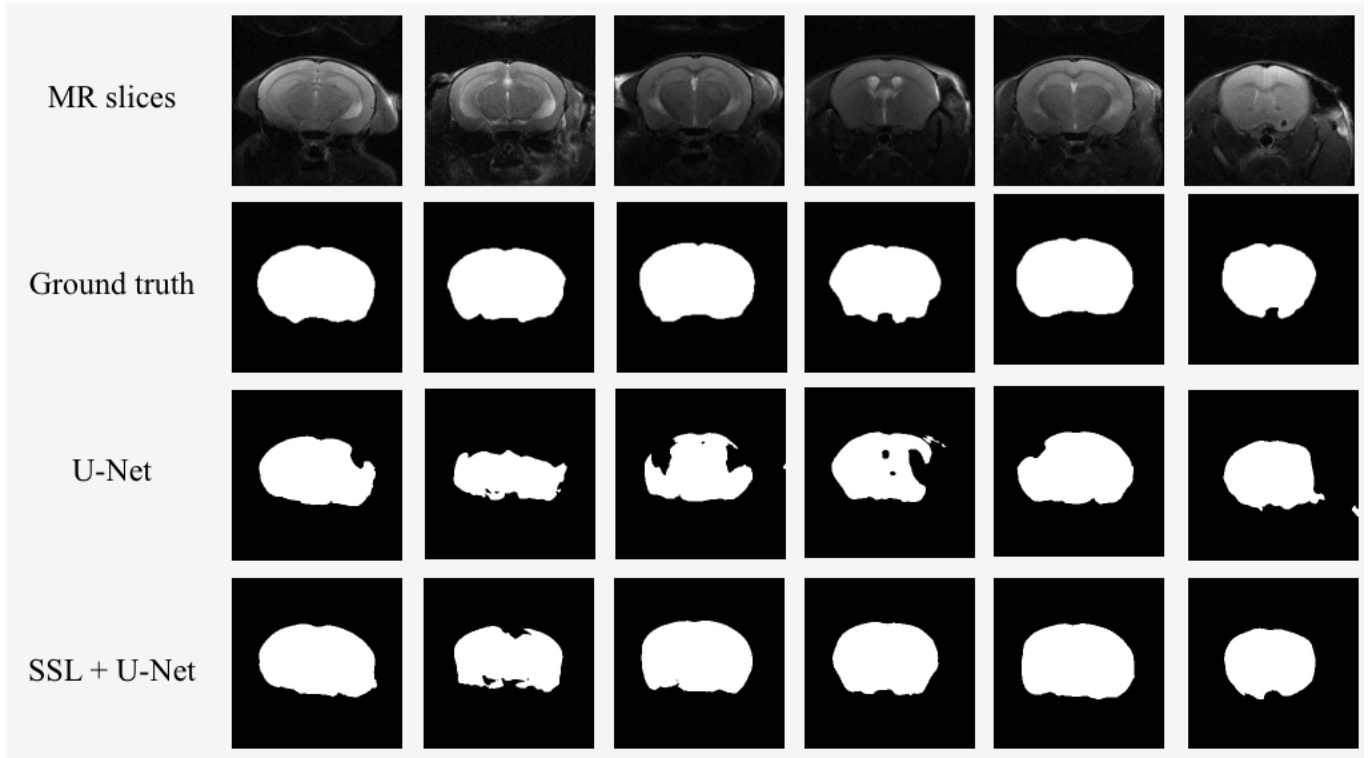


Fig. 4. Qualitative segmentation comparison between the baseline UNet and the SSL-pretrained UNet. The first row shows the original MRI slices. The second row shows manual segmentation from experts. The third and the fourth row show U-Net and the proposed two-stage segmentation results, respectively

a mean DSC of 0.80 on the test set. In contrast, the model initialized with SSL-pretrained weights significantly outperformed the baseline under the same labeling budget, improving the DSC from 0.80 to 0.88. Qualitative analysis in Fig. 4 further demonstrates that the SSL-enhanced model yields sharper boundary delineation and superior robustness. These results suggest that SSL-based pretraining provides a superior initialization, effectively mitigating performance degradation in data-constrained environments.

IV. DISCUSSION AND CONCLUSION

This study demonstrates that self-supervised learning (SSL) significantly improves 2D mouse brain MRI segmentation under limited labeled data conditions. The MIM-based pretraining provides measurable improvements over a baseline UNet trained from scratch. Specifically, SSL pretraining increased the DSC from approximately 0.80 to 0.88 while maintaining identical fine-tuning settings and labeled data volume.

Our ablation analysis further reveals that MIM performance is highly sensitive to patch size and masking ratio. High masking ratios degraded reconstruction performance because excessive information removal renders the reconstruction task ill-posed. Similarly, very small patches emphasize fine-grained textures without sufficiently enforcing global reasoning, while very large patches may remove too much contiguous structure at once. Consequently, the best-performing configuration (patch size 32×32 , masking ratio 0.5) represents an optimal balance between task difficulty and contextual learning.

Qualitative comparisons reinforce these quantitative improvements. The baseline UNet occasionally produced irregular boundaries and small false-positive regions, particularly near peripheral brain contours. In contrast, the SSL-pretrained model generated smoother and more anatomically consistent segmentations. These improvements suggest that SSL acts as an implicit regularizer, improving generalization and reducing overfitting to limited annotations. This is particularly vital in preclinical imaging studies, where acquiring additional labeled data is often cost-prohibitive and time-consuming.

Despite these promising results, several limitations remain. First, the dataset size is relatively small (21 subjects); while subject-level splitting mitigates data leakage, validation on larger, multi-site datasets is required to confirm generalizability. Second, the current experiments process 2D slices independently; incorporating 3D volumetric context could further improve anatomical coherence. Third, this study focused on a UNet backbone. Future work will explore more advanced architectures such as UNet++ [16], attention-based networks, or transformer-based encoders, which have shown promising results in MRI tasks using MIM [10]. Extending this framework to 3D segmentation with larger datasets and advanced backbones remains a priority to further enhance performance.

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